

Evaluation of the Possible Transmission of BSE and Scrapie to Gilthead Sea Bream (*Sparus aurata*)



In transmissible spongiform encephalopathies (TSEs), a group of fatal neurodegenerative disorders affecting many species, the key event in disease pathogenesis is the accumulation of an abnormal conformational isoform (PrP^{Sc}) of the host-encoded cellular prion protein (PrP^C). While the precise mechanism of the PrP^C to PrP^{Sc} conversion is not understood, it is clear that host PrP^C expression is a prerequisite for effective infectious prion propagation. Although there have been many studies on TSEs in mammalian species, little is known about TSE pathogenesis in fish. Here we show that while gilthead sea bream (*Sparus aurata*) orally challenged with brain homogenates prepared either from a BSE infected cow or from scrapie infected sheep developed no clinical prion disease, the brains of TSE-fed fish sampled two years after challenge did show signs of neurodegeneration and accumulation of deposits that reacted positively with antibodies raised against sea bream PrP. The control groups, fed with brains from uninfected animals, showed no such signs. Remarkably, the deposits developed much more rapidly and extensively in fish inoculated with BSE-infected material than in the ones challenged with the scrapie-infected brain homogenate, with numerous deposits being proteinase K-resistant. These plaque-like aggregates exhibited congophilia and birefringence in polarized light, consistent with an amyloid-like component. The neurodegeneration and abnormal deposition in the brains of fish challenged with prion, especially BSE, raises concerns about the potential risk to public health. As fish aquaculture is an economically important industry providing high protein nutrition for humans and other mammalian species, the prospect of farmed fish being contaminated with infectious mammalian PrP^{Sc}, or of a prion disease developing in farmed fish is alarming and requires further evaluation.